Morphological Classification of Galaxies in HST and JWST Images Using Classical DIP

Weekly Progress Report — Week 5

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ECE 501 – Digital Image Processing Course Project

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1 Purpose and Scope

The Week 4 progress report indicated that NGC 7318A/B might become one unified object and that Adaptive thresholds may not be met for faint tidal tails. A follow-up code audit resulted in a finding of a more basic problem: The Stephan's Quilt consists of five galaxies marked by circles, but the segmentation error resulted in a single connected galaxy.The segmentation error caused the five galaxies in the Stephan's Quilt to collapse into a single connected galaxy, as shown by the circles of approximately 4,000,000 pixels. This led to an accuracy rate of 0.0 for classification and left only one of the data points unclassified. The five galaxies catalogued that can be extracted as a cut out. This report describes the auditing process, bugs discovered and fixes. The application of the procedure, the results of verification and what still needs to be done before the classification and benchmarking at Week 5.

2 Audit Process

The notebook was reviewed in two passes:

- **Pass 1 — Initial audit:** An external bug audit identified eight issues spanning segmentation, star/galaxy separation, cutout extraction, CAS feature computation (Smoothness and Asymmetry), a detector-alignment concern in the RGB composite, and two low-severity code issues. Each claim was checked directly against the notebook code before being accepted.
- **Pass 2 — Follow-up review:** After the first set of fixes was applied, a second review of the follow-up edit (`ECE501_1.ipynb`) found that the segmentation fix had been structurally undone. The watershed split was computed for a comparison figure, but the actual pipeline mask was rebuilt from the original single adaptive threshold. This was corrected so that split cores flow through morphological cleanup, star/galaxy separation, and cutout extraction.

Two items from the first pass could not be independently confirmed because the raw FITS data and headers were unavailable: the precise WCS/`NoConvergence` failure mode behind the single-cutout issue, and a possible NRCB1/NRCB2 detector mismatch in the RGB composite. These remain plausible but unverified.

3 Bugs Identified and Fixes Applied

The code audit identified eight issues. The causes and corrective measures are summarised in Table 1 below. The fixes were tested against the notebook program - this does not mean that every fixed output has been regenerated compared the calculated results with the astronomical data.

Table 1: Bugs identified during the audit and their remediation.

Bug	Root cause	Fix applied	Status
Segmentation collapses into one giant component	OpenCV's adaptive threshold convention is $T(x,y) = \text{mean}_{\text{local}} - C$. The offset sign was wrong, lowering the local threshold below the sky median. Closing and hole-filling then bridged the flooded region into a roughly 4,000,000-pixel blob.	Replaced single-threshold segmentation with an Otsu foreground footprint and distance-transform watershed. Cleanup (closing, opening, and hole-filling) is applied per watershed label so adjacent cores cannot re-merge.	Fixed
JWST classification accuracy measured at 0.0%	With only one segmented object, top-N area ranking forced a single blob onto one ground-truth catalog member, guaranteeing a mismatch.	Addressed downstream of the segmentation correction: multiple distinguishable galaxy candidates can now be produced, allowing accuracy to be computed over more than one match.	Fixed downstream of Bug 1
Only NGC 7317 extracted as a cutout	Only one candidate blob was available. The precise WCS/astrometric matching failure could not be verified without the raw FITS headers.	Restored multiple segmented cores and hardened candidate matching by pre-computing topological reference centres rather than recomputing them inside a shrinking loop.	Fixed structurally
Smoothness (S) inflated by boundary spillover	Gaussian blur was applied to a cutout zeroed outside the galaxy mask. Flux spilled across the hard mask edge, creating a boundary residual proportional to perimeter rather than genuine structure.	Blur is applied to the unmasked cutout (or a median-filled background), and the $\lvert I - I_{\text{smooth}}\rvert$ residual is summed only within the mask aperture.	Fixed
Asymmetry (A) background correction used mismatched scales	A_{bg} was computed from raw, unstretched FITS flux (MJy/sr), while A_{raw} used 8-bit asinh-stretched, background-subtracted cutouts.	The sky patch is now taken from the same 8-bit stretched array as the cutouts, placing both terms on the same scale before subtraction.	Fixed
Hardcoded sky patch not guaranteed to be empty sky	A fixed `raw_f200[100:300,100:300]` patch was used for SNR checks and CAS background correction, although it may overlap a source depending on the frame or crop.	A dynamic corner selector evaluates all four frame corners and chooses the one furthest from the flux-weighted bright-source centroid.	Fixed
Concentration (C) inconsistent across weeks	Week 3 used an inner-flux/total-flux fraction with range 0–1 and threshold 0.25; Week 4 used the Conselice curve-of-growth definition, with a typical range around 2–5. The scales were not comparable.	Standardised both weeks on $C = 5\log_{10}(r_{80}/r_{20})$ and updated `C_THRESH` from 0.25 to 2.5.	Fixed
Plotting label typo and division-by-zero risk	A Gaussian-fit legend omitted equals signs, and the class-distribution printout divided by the detection count without checking for zero detections.	Corrected the label and added a zero-count guard that prints 0.0% when there are no detections.	Fixed

4 Final Week 4 Pipeline Configuration

Table 2 lists the corrected configuration of the pipeline. The adaptive threshold parameters are still available for the E1 comparison visualisation, but the adaptive threshold is no longer providing downstream segmentation mask.

Table 2: Final configuration recorded after the audit.

Parameter	Final value	Notes
Segmentation method	Otsu + distance-transform watershed	Replaces the single adaptive-threshold segmentation.
Adaptive threshold block size	`W4_SEG_BLOCK_SIZE = 51` px	Retained for E1 visual comparison only.
Adaptive threshold offset	`W4_SEG_C_OFFSET = -5`	No longer drives the downstream segmentation mask.
Morphological cleanup	5×5 elliptical kernel; per-label closing (2 iterations), opening (1 iteration), and hole-fill	Applied per watershed label to prevent adjacent cores from re-merging.
Minimum object area	`W4_SEG_MIN_AREA_PX = 200` px	Unchanged.
Star/galaxy concentration threshold	`W4_SGS_CONC_THRESH = 0.35`	Uses a circular, flux-weighted aperture rather than a square inner box.
Star/galaxy area threshold	`W4_SGS_AREA_MAX_PX = 1500` px	Unchanged.
Concentration definition	$C = 5 \log_{10}(r_{80}/r_{20})$	Conselice (2003) form standardised across Weeks 3 and 4.
Concentration threshold	`C_THRESH = 2.5`	Updated from 0.25 to match the Conselice scale.
Asymmetry background patch	Dynamic corner of `sq_u8` (8-bit stretched scale)	Corner selected by distance from the source centroid.
Smoothness background patch	Same dynamic corner, median-filled before blur	Reduces boundary spillover into zero-valued background.

5 Results Obtained

These outcomes were directly verified by the corrected code and the controlled synthetic tests:

- A synthetic two-core test substantiated the cleanup on per-label. Two neighbouring labeled regions, sufficiently near each other, such that a they were separate following the full per-label closing, opening, and hole-filling sequence.
- Each of the edited notebook cells was independently read after every round of edits by the editor to check that the code that they wanted was provided, not a partial or out-of-date version.
- After each set of editing, the notebook was checked with nbformat to be structurally correct.

There was no actual execution of the real JWST/HST FITS data. The raw mastDownload/ files and some packages (astropy, photutils, astroquery) were not available in the audit environment. Therefore, number of connected component found, the number of member of Stephan's quintet found and extracted and classification accuracy is unregenerated. At this time no new numerical results are claimed in this report.

6 Planned Verification Before Week 5

The following checks remain necessary before proceeding with the Week 5 rule-based classification and Galaxy Zoo benchmarking:

1. Re-execute the corrected notebook using *Restart & Run All* against the real FITS data. Record the actual post-fix connected-component count, number of matched Stephan's Quintet cutouts, and classification accuracy, replacing the placeholder figures in the Week 4 report.
2. Independently check the WCS/`NoConvergence` cutout-matching behaviour and the NRCB1/NRCB2 detector-alignment question using the actual FITS headers.

3. Tune the watershed `min_distance` parameter against the real image if the recovered number of cores under- or over-splits relative to the five known Stephan's Quintet members.
4. Carry the corrected and re-executed CAS feature table into the Week 5 rule-based classification and Galaxy Zoo benchmarking stage.